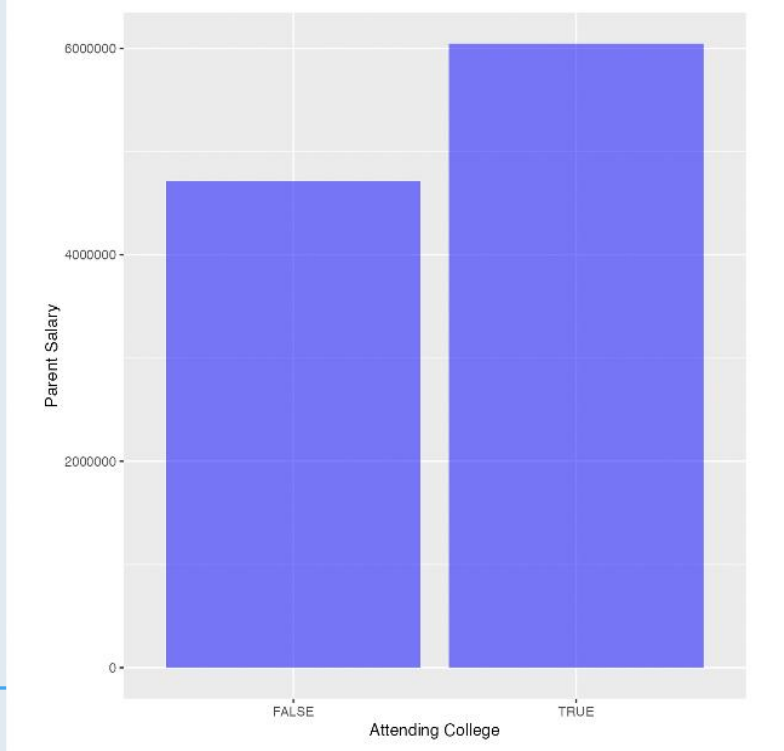
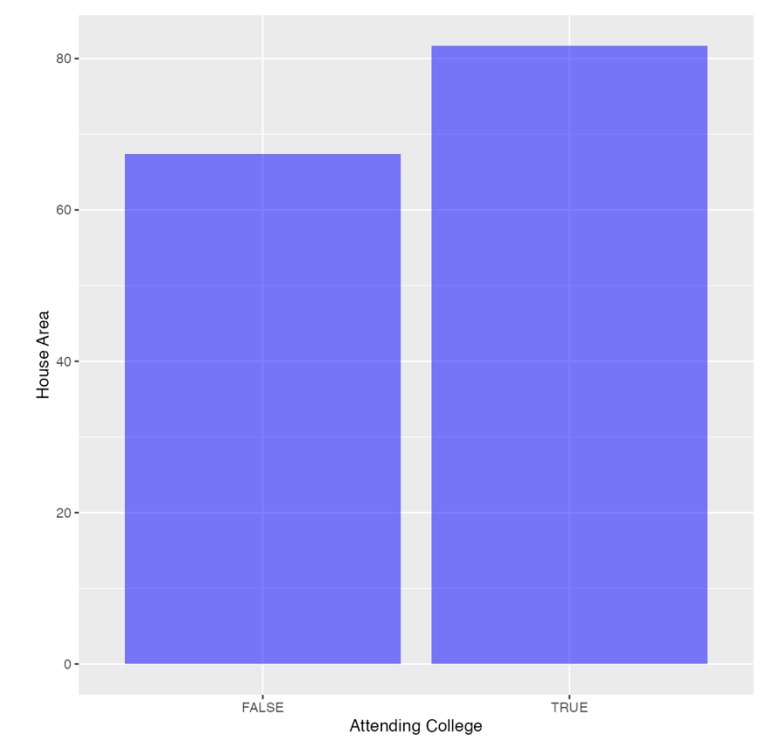
FACTORS AFFECTING COLLEGE ATTENDANCE

NATHAN GEBREMICHAEL

Introduction

- Dataset created by National Institute of Statistics and Geography, sampled 27,000 Indonesian high school students.
- Although Indonesia does not have a particular problem with college attendance (Trading Economics), we were interested in whether this dataset reflected the global trends of how socioeconomic status and privilege are related to college attendance (Reber and Smith).
- Given a college degree's widespread influence on people's ability to bring in sustainable income, privilege gatekeeping college attendance continues to be a relevant problem that we felt could be explored through this focus.





Research Questions/Model

- Main research question: What variables related to socioeconomic status/privilege are predictors of a high school student's decision to attend college?
  - Can a student's average grades be tied to socioeconomic status?
- If socioeconomic status is a predictor of college attendance, can we make any recommendations based on our model to address these predictors' influence?

```
#Recode will_go_to_college as binary

college_motivation$will_go_to_college_binary <- ifelse(college_motivation$will_go_to_college == "True", 1, 0)

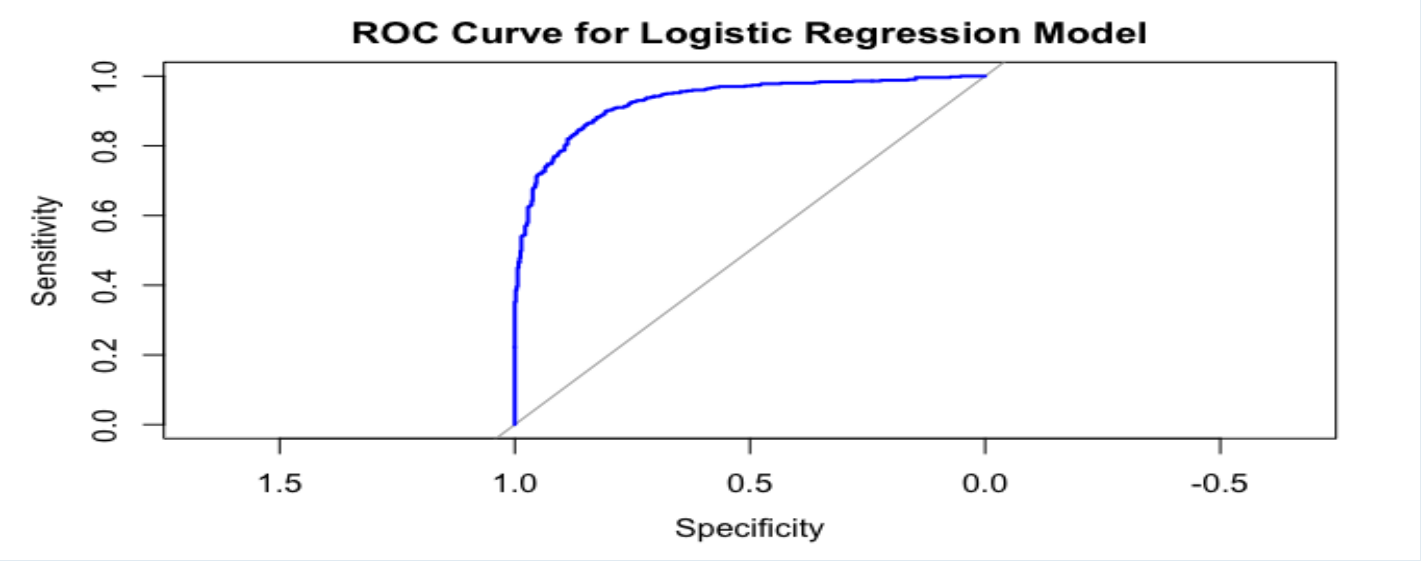
View(college_motivation)

#Model!

college_motivation_model <- glm(will_go_to_college_binary ~ gender + residence + parent_salary + house_area + average_grades, data = college_motivation, family = "binomial")
summary(college_motivation_model)
```

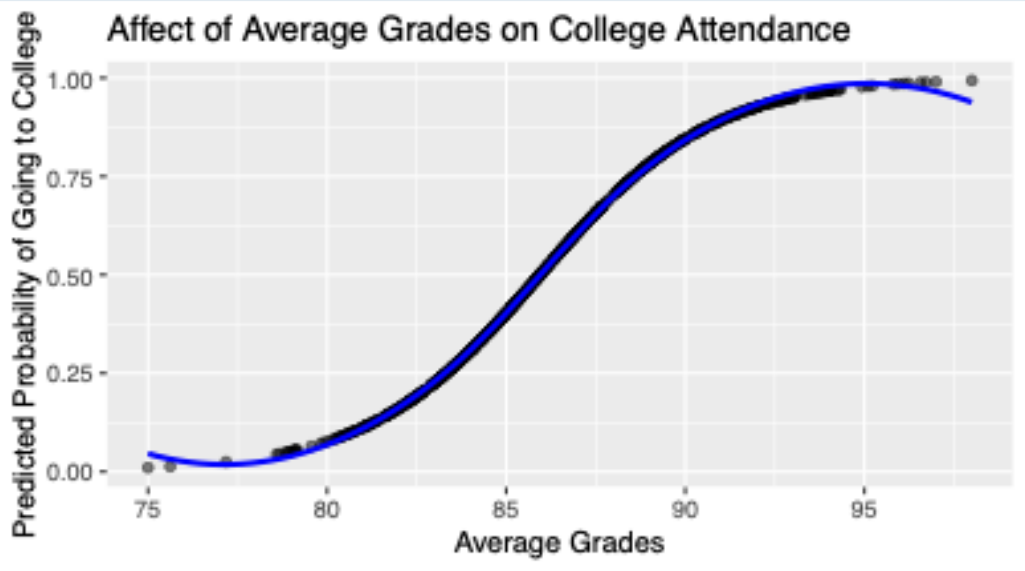
Model Results

Variable	Estimate	Std. Error	z-value	p-value
Intercept	-44.249	3.298	-13.417	<2E-16
Gender	-0.254	0.194	-1.306	0.192
Residence	2.623	0.259	10.134	<2E-16
Parent Salary	1.22E-06	9.72E-08	12.54	<2E-16
House Area	0.103	0.009	11.176	<2E-16
Average Grades	0.333	0.034	9.665	<2E-16



- PMSE value of model: 0.147 (Quite accurate!)
- AUC-ROC value: 0.929 (Very strong)

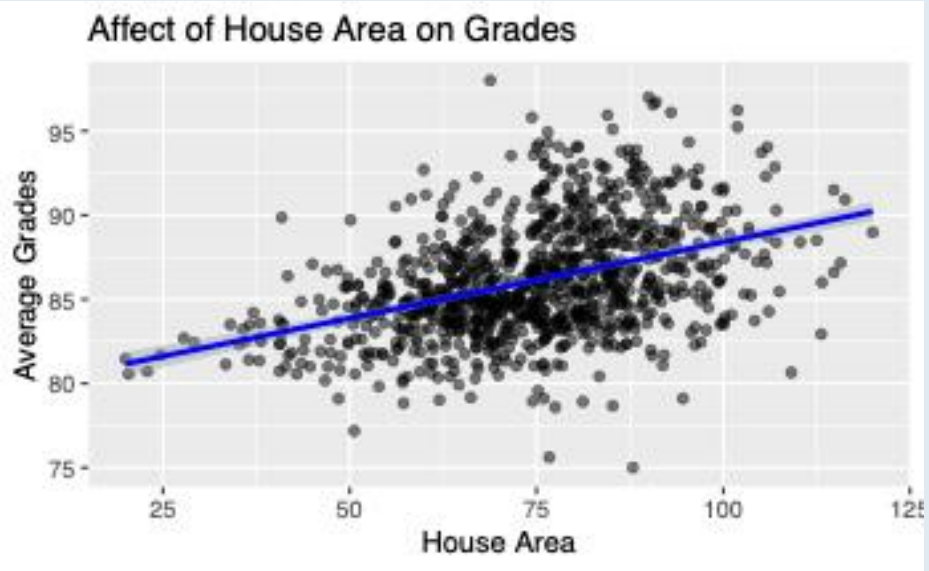
Do Average Grades Affect Going to College?

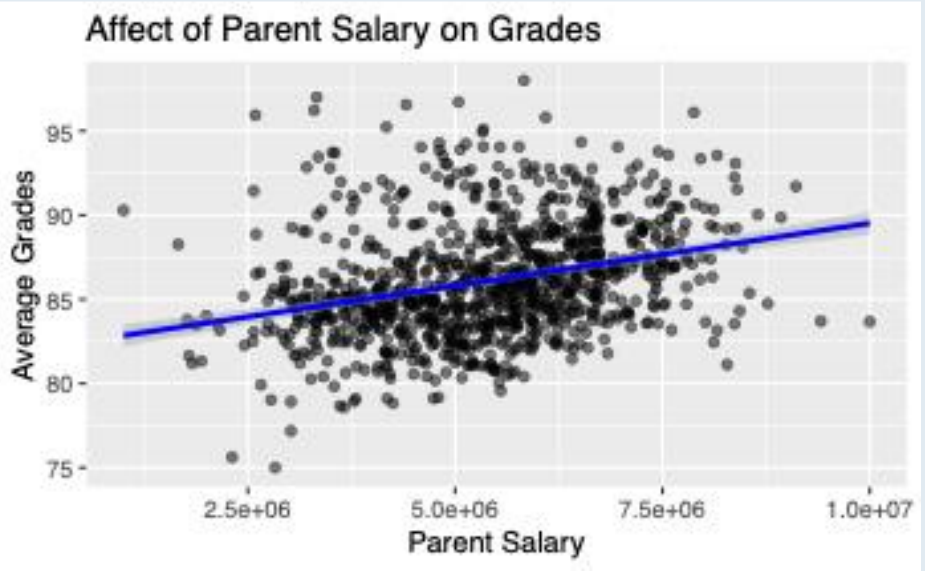


Variable	Estimate	Std. Error	z-value	p-value
Intercept	-35.974	2.586	-13.91	<2e-16
Average Grades	0.419	0.03	13.89	<2e-16

- Statistically significant p-value (<2e-16) indicates that average grades effects the likelihood of going to college

What Factors Affect Average Grades?





Variable	Estimate	Std. Error	z-value	p-value
Intercept	7.33E+01	1.719	42.619	<2e-16
Urban Residence	-7.23E-01	2.38E-01	-3.043	0.002
Parent's Age	8.32E-02	3.34E-02	2.494	0.013
Parent's Salary	6.11E-07	7.15E-08	8.547	<2e-16
House Area (m^2)	7.50E-02	6.62E-03	11.339	<2e-16

- Parent's salary and house area had the greatest affect on average grades
  - Both had statistically significant p-values

USA Comparison

- "The report indicates that in 2022, an estimated 79 percent of 18-to-24-year-olds from the highest family income quartile enrolled in postsecondary education, compared with just 44 percent from the lowest income quartile. Historical data suggests that higher levels of educational attainment are associated with higher incomes and greater job opportunities" (Hamm, 2024).

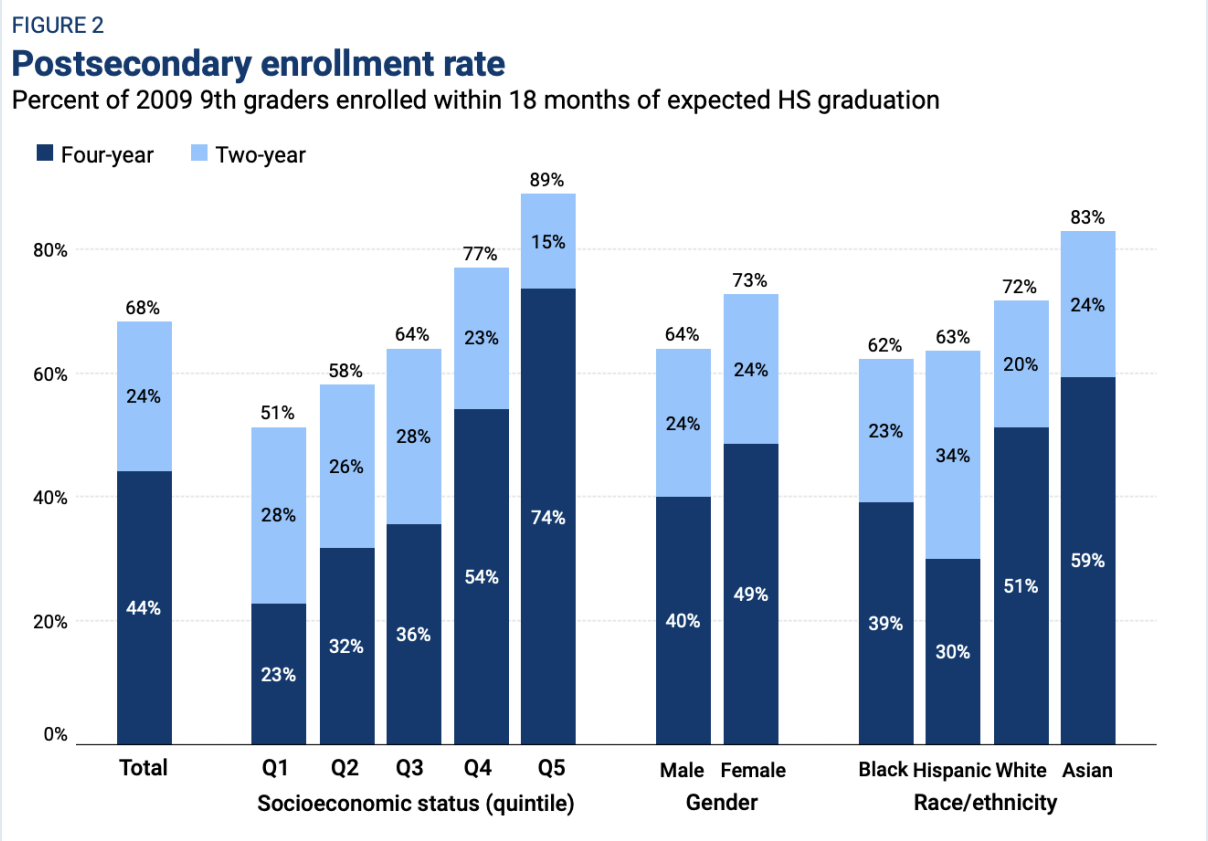


Fig. 2. Postsecondary enrollment rate by Sarah Reber and Ember Smith; "College Enrollment Disparities"; *Brookings.edu*, 04 Dec. 2024, https://www.brookings.edu/wp-content/uploads/2023/02/20230123_CCF_CollegeEnrollment_FINAL2.pdf

Discussion/Interpretation of Results

- Socioeconomic status and privilege are clearly influential on Indonesian high school students' decision to attend college. This reflects general global trends, and more specifically, trends here in the United States.
- Parent salary- Recommend more widely available aid for students, student loan forgiveness, reduced tuition and fees
- Average grades- related to SES and privilege. Recommend implementing extra help for students that controls for these factors
- Every predictor *besides* gender was a significant predictor of college attendance. Why is this surprising?
- Unfinished ideas:
    - Trying other models - variable selection, test/train, etc
    - Incorporating a second raw data set
    - Race/Ethnicity as factors related to privilege

Sources

- Hexter, Holly, et al. “New Pell Institute Report Shows Decline in the Global Position of the United States in Bachelor’s Attainment, Increasing Inequality of College Opportunity at Home.” *Council for Opportunity in Education*, 10 May 2024, coenet.org/news-impact/press-releases/new-pell-institute-report-shows-decline-in-the-global-position-of-the-united-states-in-bachelors-attainment-increasing-inequality-of-college-opportunity-at-home/#:~:text=The%20report%20indicates%20that%20in,from%20the%20lowest%20income%20quartile.

- "Indonesia School Enrollment, Tertiary (Percent of Gross) - World Bank Data." Trading Economics, Trading Economics, www.tradingeconomics.com/indonesia/school-enrollment-tertiary-percent-gross-wb-data.html. Accessed 6 Dec. 2024.

- Reber, Sarah, and Ember Smith. “College Enrollment Disparities.” *COLLEGE ENROLLMENT DISPARITIES*, Brookings, www.brookings.edu/wp-content/uploads/2023/02/20230123_CCF_CollegeEnrollment_FINAL2.pdf. Accessed 4 Dec. 2024.